Theorem 6.5.2 (Gabriel's theorem)**.** *Let Q be a quiver of type A_n, D_n, E_6, E_7, E_8. Then Q has finitely many indecomposable representations. Namely, the dimension vector of any indecomposable representation is a positive root (with respect to B_Γ) and for any positive root α there is exactly one indecomposable representation with dimension vector α.*

6.6. Reflection functors

Definition 6.6.1. Let Q be any quiver. We call a vertex $i \in Q$ a **sink** if all edges connected to i point towards i:

$$\begin{tikzcd} & i \arrow[dl] & \\ \bullet \arrow[r] & \bullet & \bullet \arrow[l] \end{tikzcd} .$$

We call a vertex $i \in Q$ a **source** if all edges connected to i point away from i:

$$\begin{tikzcd} & i \arrow[dr] & \\ \bullet & \bullet \arrow[l] \arrow[r] & \bullet \\ & \bullet \arrow[u] & \end{tikzcd} .$$

Definition 6.6.2. Let Q be any quiver and let $i \in Q$ be a sink (respectively, a source). Then we let $\overline{Q_i}$ be the quiver obtained from Q by reversing all arrows pointing into (respectively, pointing out of) i.

We are now able to define the **reflection functors** (also called **Coxeter functors**).

Definition 6.6.3. Let Q be a quiver, and let $i \in Q$ be a sink. Let V be a representation of Q. Then we define the reflection functor

$$F_i^+ : \operatorname{Rep} Q \to \operatorname{Rep} \overline{Q_i}$$

by the rule

$$\begin{aligned} F_i^+(V)_k &= V_k \quad \text{if } k \neq i, \\ F_i^+(V)_i &= \ker\left(\varphi : \bigoplus_{j \to i} V_j \to V_i\right). \end{aligned}$$